

GED Science Practice Test 4

PART A — QUESTIONS

GED-SCI4-001

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A class investigates whether the amount of dissolved fertilizer affects duckweed growth. Four containers receive the same amount of pond water, the same starting number of duckweed plants, and the same light exposure. Each container receives a different fertilizer concentration. After 7 days, students count the number of duckweed plants.

Fertilizer concentration (mg/L): 0, 5, 10, 20

Duckweed plants after 7 days: 12, 21, 35, 28

Which conclusion is best supported by the data?

- A) Duckweed grew most with 10 mg/L fertilizer.
- B) Duckweed grew most with no fertilizer.
- C) Fertilizer concentration had no effect on duckweed growth.
- D) The highest fertilizer concentration always produced the most growth.

GED-SCI4-002

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student tests four unknown liquids by measuring mass and volume.

Liquid	Mass (g)	Volume (mL)
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A	36	30
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B	42	20
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C	25	25
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D	54	18
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Which liquid has a density of 1.0 g/mL?

- A) Liquid A
- B) Liquid B
- C) Liquid C
- D) Liquid D

GED-SCI4-003

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A scientist explains that most earthquakes occur near plate boundaries. At one type of boundary, plates move apart and magma rises to form new crust. This process can form mid-ocean ridges on the seafloor.

Which type of plate boundary is described?

- A) Divergent boundary
- B) Transform boundary
- C) Convergent boundary
- D) Passive boundary

GED-SCI4-004

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Red blood cells contain hemoglobin, a protein that binds oxygen. When blood passes through the lungs, oxygen attaches to hemoglobin. The blood then carries oxygen to body tissues, where cells use oxygen during cellular respiration.

Which statement best describes the main function of red blood cells in this passage?

- A) They carry oxygen from the lungs to body tissues.
- B) They digest food in the small intestine.
- C) They send electrical signals to muscles.
- D) They filter wastes from the blood.

GED-SCI4-005

Section: Science

Type: numeric_entry

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A sample of a liquid has a mass of 75 g and a volume of 30 mL.

What is the density of the liquid in g/mL?

Type your answer as a decimal rounded to the nearest tenth.

GED-SCI4-006

Section: Science

Type: multi_select

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student wants to test whether wire thickness affects the resistance of a circuit. The student uses copper wires of different thicknesses but keeps the wire length, battery, and bulb the same.

Which TWO factors are controlled variables in this experiment?

- A) Wire length

- B) Battery used
- C) Wire thickness
- D) Resistance measured
- E) Brightness result

GED-SCI4-007

Section: Science

Type: dropdown

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A cell contains ribosomes that build proteins using instructions from genetic material. In this sentence, ribosomes are mainly involved in [D1], while genetic material mainly stores [D2].

Dropdowns:

D1) protein production | water storage | gas exchange

D2) inherited instructions | thermal energy | dissolved minerals

GED-SCI4-008

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student compares how quickly equal masses of powdered sugar and sugar cubes dissolve in water. Both samples are placed into 150 mL of water at 25°C and stirred at the same speed.

What is the independent variable?

- A) Form of the sugar
- B) Volume of water
- C) Water temperature
- D) Stirring speed

GED-SCI4-009

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A farmer notices that after a long period without rain, soil becomes dry and loose. When strong winds blow across the field, small soil particles are carried away. Planting cover crops can help hold soil in place.

Which process is mainly described?

- A) Wind erosion
- B) Condensation
- C) Subduction
- D) Solar fusion

GED-SCI4-010

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A nurse records blood glucose levels for a patient before and after a meal.

Time | Blood glucose (mg/dL)

Before meal | 88

30 min after meal | 142

60 min after meal | 126

120 min after meal | 94

Which statement is best supported by the data?

- A) Blood glucose rose after the meal and then moved back toward the starting level.
- B) Blood glucose stayed exactly the same for two hours.
- C) Blood glucose was lowest 30 minutes after the meal.
- D) Blood glucose increased continuously during the entire period.

GED-SCI4-011

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student wants to determine whether a coating reduces rusting on steel nails. Ten coated nails and ten uncoated nails are placed in identical cups of salt water. After five days, the student records how many nails show rust.

Why are uncoated nails included?

- A) They provide a comparison group for the coated nails.
- B) They make the salt water less reactive.
- C) They prevent coated nails from rusting.
- D) They change the coating into steel.

GED-SCI4-012

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

During digestion, enzymes break large food molecules into smaller molecules. The small intestine absorbs many of these smaller molecules into the blood. The circulatory system then transports them to cells throughout the body.

Which two body systems are working together most directly in this process?

- A) Digestive and circulatory systems
- B) Nervous and skeletal systems
- C) Respiratory and reproductive systems
- D) Integumentary and muscular systems

GED-SCI4-013

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student places a warm metal rod in a cup of cooler water. After several minutes, the metal rod is cooler and the water is warmer.

Which statement best explains the energy transfer?

- A) Thermal energy moved from the warmer metal rod to the cooler water.
- B) Cold moved from the water into the metal rod.
- C) The metal rod destroyed thermal energy.
- D) Thermal energy moved only if the objects had the same temperature.

GED-SCI4-014

Section: Science

Type: numeric_entry

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A stream carries sediment 36 kilometers over 9 days during a flood season.

What is the average distance the sediment moves per day?

Type your answer as a whole number.

GED-SCI4-015

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Researchers test whether a new handwashing procedure reduces the spread of a harmless classroom virus model made from glowing powder. One group follows the new procedure, while another group washes hands as usual. After handwashing, researchers count glowing spots left on students' hands.

Which evidence would best support the new procedure?

- A) The new-procedure group has fewer glowing spots on average.
- B) The usual-washing group has fewer glowing spots on average.
- C) Both groups have identical numbers of glowing spots.
- D) The glowing powder is counted before any handwashing occurs.

GED-SCI4-016

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A bottle of soda is opened, and bubbles rise through the liquid. The bubbles are carbon dioxide gas that was dissolved in the soda under pressure. When the cap is removed, the pressure decreases and gas leaves the liquid.

Which change occurs when the soda is opened?

- A) Dissolved CO₂ leaves the liquid as gas bubbles.
- B) CO₂ changes into solid salt.
- C) The liquid gains unlimited mass.
- D) The gas disappears because gases have no volume.

GED-SCI4-017

Section: Science

Type: dropdown

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

In a simple circuit, a closed switch allows electric current to flow. An open switch [D1] the circuit, so the bulb [D2].

Dropdowns:

D1) breaks | strengthens | cools

D2) turns off | becomes a battery | produces water

GED-SCI4-018

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student tests whether different music volumes affect the rate at which crickets chirp. The student places crickets in identical containers, keeps the temperature the same, and plays sound at different volumes. The number of chirps per minute is recorded.

Why should temperature be kept the same?

- A) Temperature can affect chirping, so keeping it constant helps isolate the effect of volume.
- B) Temperature is the dependent variable in every animal experiment.
- C) Keeping temperature constant guarantees all crickets chirp at the same rate.
- D) Temperature must change whenever sound volume changes.

GED-SCI4-019

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures the height of a ball dropped from rest every 0.5 second.

Time (s): 0, 0.5, 1.0, 1.5

Height above ground (m): 20, 18.8, 15.1, 8.9

Which statement is best supported by the data?

- A) The ball moves downward and covers more distance during each later interval.
- B) The ball remains at the same height the whole time.
- C) The ball moves upward at a constant speed.
- D) The ball gains height as time increases.

GED-SCI4-020

Section: Science

Type: multi_select

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

Select TWO answers.

A town records summer water use and average reservoir level.

Year	Summer water use (million L)	Average reservoir level (% full)
1	180	82
2	205	74
3	230	66
4	250	58

Which TWO statements are best supported by the data?

A) Summer water use increased over the four years.

B) Average reservoir level decreased over the four years.

C) Reservoir level increased as water use increased.

D) Water use stayed the same each year.

E) The reservoir was fullest in Year 4.

GED-SCI4-021

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A population of wild rabbits lives in an area where winters become warmer over many years.

Rabbits with thinner coats survive as well as or better than rabbits with very thick coats. Over many generations, thinner coats become more common.

Which explanation best fits this change?

A) Rabbits with thinner coats had a survival or reproductive advantage in the warmer environment.

B) Thick-coated rabbits chose to grow thinner coats during their lifetime.

C) The environment stopped influencing rabbit survival.

D) All rabbits became genetically identical after one winter.

GED-SCI4-022

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student claims that increasing water temperature increases the rate of dissolving salt. The student adds 5 g of salt to 100 mL of water at three temperatures and stirs each cup at the same speed.

Water temperature (°C)	10	30	50
Time to dissolve (s)	95	61	34

Which statement best evaluates the claim?

- A) The data support the claim because salt dissolved faster at higher temperatures.
- B) The data do not support the claim because the dissolving time increased.
- C) The data prove that temperature has no effect on dissolving.
- D) The data are unrelated because the same amount of salt was used.

GED-SCI4-023

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A text model of the rock cycle describes these steps: magma cools and solidifies to form igneous rock; weathering breaks rock into sediment; sediment can be compacted and cemented into sedimentary rock; heat and pressure can change existing rock into metamorphic rock.

Which process forms sedimentary rock in this model?

- A) Compaction and cementation of sediment
- B) Cooling of magma at Earth's surface
- C) Melting of rock into magma
- D) Movement of planets around the Sun

GED-SCI4-024

Section: Science

Type: numeric_entry

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

In a study of 160 fruit flies, 40 have white eyes.

What percent of the fruit flies have white eyes?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI4-025

Section: Science

Type: drag_drop

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Match each target with the correct tile.

Tiles:

- A) Mass added to the cart
- B) Acceleration of the cart
- C) Same track surface for all trials
- D) Color of the cart

Targets:

- T1) Independent variable
- T2) Dependent variable
- T3) Controlled variable

GED-SCI4-026

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A mutation changes the DNA sequence of a gene in a skin cell. The mutation is not present in sperm or egg cells. The skin cell may pass the mutation to new skin cells when it divides, but the mutation will not be passed to the person's children.

Why will this mutation not be inherited by the person's children?

- A) The mutation is not in reproductive cells.
- B) Skin cells do not contain any DNA.
- C) All mutations immediately disappear.
- D) Children inherit only proteins, not genetic information.

GED-SCI4-027

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student records the temperature of water being heated.

Time (min): 0, 3, 6, 9

Temperature (°C): 18, 30, 42, 54

What is the average rate of temperature increase from 0 to 9 minutes?

- A) 2°C/min
- B) 3°C/min
- C) 4°C/min
- D) 6°C/min

GED-SCI4-028

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

Earth rotates once about every 24 hours. As Earth rotates, different locations move into and out of sunlight.

Which observation is mainly caused by Earth's rotation?

- A) Day and night
- B) Seasons
- C) Ocean tides only
- D) Formation of sedimentary rock

GED-SCI4-029

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A person eats a meal containing carbohydrates. Digestive enzymes break many carbohydrates into glucose. Body cells use glucose during cellular respiration to release usable energy.

Which statement best describes the role of glucose in body cells?

- A) Glucose is a fuel source used to release energy.
- B) Glucose carries oxygen in red blood cells.
- C) Glucose forms bones by replacing calcium.
- D) Glucose prevents all diseases.

GED-SCI4-030

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares the energy needed to warm different samples of water.

Sample	Mass of water (g)	Temperature increase (°C)	Energy added (J)
A	50	10	2,100
B	100	10	4,200
C	150	10	6,300
D	200	10	8,400

Which statement is best supported by the table?

- A) More energy is needed to produce the same temperature increase in a larger mass of water.
- B) Less energy is needed as the mass of water increases.
- C) Energy added is unrelated to the mass of water.
- D) The temperature increase is greatest for Sample D.

GED-SCI4-031

Section: Science

Type: multi_select

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student studies whether hand temperature affects pulse rate after mild exercise. The student measures pulse after the same exercise while hands are warmed, cooled, or kept at room temperature.

Which TWO steps would improve the investigation?

- A) Use the same exercise length before each pulse measurement.
- B) Measure pulse for the same amount of time in each trial.
- C) Change the exercise type before every trial.
- D) Record only the highest pulse from all trials.
- E) Use a different pulse-counting method each time.

GED-SCI4-032

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares the speed of pulses traveling through different ropes.

Rope | Distance traveled (m) | Time (s)

A | 12 | 3

B | 20 | 4

C | 18 | 6

D | 10 | 5

Which rope has the fastest pulse speed?

A) Rope A

B) Rope B

C) Rope C

D) Rope D

GED-SCI4-033

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A forest food web includes oak trees, caterpillars, small birds, hawks, and fungi. Oak trees make sugars using sunlight. Caterpillars eat oak leaves, small birds eat caterpillars, and fungi break down dead organisms and return nutrients to the soil.

What is the role of fungi in this food web?

A) Decomposers

B) Producers

C) Primary consumers

D) Top predators

GED-SCI4-034

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A vaccine contains a harmless piece of a virus. After vaccination, the immune system makes antibodies and memory cells. Months later, if the real virus enters the body, the immune system can respond more quickly.

Which conclusion is best supported?

A) Vaccination can prepare the immune system for a faster future response.

B) Vaccination replaces the need for all white blood cells.

C) Vaccination makes every virus harmless immediately.

D) Vaccination prevents the body from producing antibodies.

PART B — ANSWER KEY + EXPLANATIONS

GED-SCI4-001 — Correct: A — Correct Answer: Duckweed grew most with 10 mg/L fertilizer. — Explanation: The highest final plant count was 35 plants at 10 mg/L fertilizer.

GED-SCI4-002 — Correct: C — Correct Answer: Liquid C — Explanation: Density = mass ÷ volume. Liquid C has density $25 \text{ g} \div 25 \text{ mL} = 1.0 \text{ g/mL}$.

GED-SCI4-003 — Correct: A — Correct Answer: Divergent boundary — Explanation: At divergent boundaries, plates move apart and new crust can form as magma rises.

GED-SCI4-004 — Correct: A — Correct Answer: They carry oxygen from the lungs to body tissues. — Explanation: Hemoglobin in red blood cells binds oxygen and helps transport it through the blood.

GED-SCI4-005 — Correct: 2.5 — Tolerance: exact — Correct Answer: 2.5 — Explanation: Density = $75 \text{ g} \div 30 \text{ mL} = 2.5 \text{ g/mL}$.

GED-SCI4-006 — Correct: A,B — Correct Answer: Wire length | Battery used — Explanation: Wire length and battery type are kept the same so wire thickness is the main variable being tested.

GED-SCI4-007 — Correct: D1=protein production; D2=inherited instructions — Correct Answer: D1=protein production; D2=inherited instructions — Explanation: Ribosomes build proteins, while genetic material stores inherited instructions.

GED-SCI4-008 — Correct: A — Correct Answer: Form of the sugar — Explanation: The form of the sugar is intentionally changed, while water volume, temperature, and stirring speed are kept constant.

GED-SCI4-009 — Correct: A — Correct Answer: Wind erosion — Explanation: Wind erosion occurs when moving air carries away soil or sediment particles.

GED-SCI4-010 — Correct: A — Correct Answer: Blood glucose rose after the meal and then moved back toward the starting level. — Explanation: Blood glucose increased from 88 to 142 mg/dL, then decreased toward 88 mg/dL.

GED-SCI4-011 — Correct: A — Correct Answer: They provide a comparison group for the coated nails. — Explanation: The uncoated nails act as a comparison group for evaluating whether the coating reduces rust.

GED-SCI4-012 — Correct: A — Correct Answer: Digestive and circulatory systems — Explanation: The digestive system breaks down and absorbs nutrients, and the circulatory system transports them.

GED-SCI4-013 — Correct: A — Correct Answer: Thermal energy moved from the warmer metal rod to the cooler water. — Explanation: Thermal energy naturally transfers from warmer objects to cooler objects.

GED-SCI4-014 — Correct: 4 — Tolerance: exact — Correct Answer: 4 — Explanation: Average distance per day = $36 \text{ km} \div 9 \text{ days} = 4 \text{ km/day}$.

GED-SCI4-015 — Correct: A — Correct Answer: The new-procedure group has fewer glowing spots on average. — Explanation: Fewer glowing spots would show that the procedure removed more of the virus model.

GED-SCI4-016 — Correct: A — Correct Answer: Dissolved CO₂ leaves the liquid as gas bubbles. — Explanation: Lower pressure after opening the bottle allows dissolved carbon dioxide to come out of solution.

GED-SCI4-017 — Correct: D1=breaks; D2=turns off — Correct Answer: D1=breaks; D2=turns off — Explanation: An open switch breaks the circuit, preventing current from flowing through the bulb.

GED-SCI4-018 — Correct: A — Correct Answer: Temperature can affect chirping, so keeping it constant helps isolate the effect of volume. — Explanation: Holding temperature constant helps determine whether volume, not temperature, affects chirp rate.

GED-SCI4-019 — Correct: A — Correct Answer: The ball moves downward and covers more distance during each later interval. — Explanation: Height decreases, and the distance fallen during each 0.5-second interval increases.

GED-SCI4-020 — Correct: A,B — Correct Answer: Summer water use increased over the four years. | Average reservoir level decreased over the four years. — Explanation: Water use rose from 180 to 250 million L, while reservoir level fell from 82% to 58%.

GED-SCI4-021 — Correct: A — Correct Answer: Rabbits with thinner coats had a survival or reproductive advantage in the warmer environment. — Explanation: Natural selection can make traits more common when they improve survival or reproduction in a specific environment.

GED-SCI4-022 — Correct: A — Correct Answer: The data support the claim because salt dissolved faster at higher temperatures. — Explanation: The time to dissolve decreased from 95 seconds to 34 seconds as temperature increased.

GED-SCI4-023 — Correct: A — Correct Answer: Compaction and cementation of sediment — Explanation: Sedimentary rock forms when sediment is compacted and cemented together.

GED-SCI4-024 — Correct: 25 — Tolerance: exact — Correct Answer: 25 — Explanation: $40 \div 160 \times 100 = 25\%$.

GED-SCI4-025 — Correct: T1=A; T2=B; T3=C — Correct Answer: T1=Mass added to the cart; T2=Acceleration of the cart; T3=Same track surface for all trials — Explanation: The independent variable is changed, the dependent variable is measured, and controlled variables are kept the same.

GED-SCI4-026 — Correct: A — Correct Answer: The mutation is not in reproductive cells. — Explanation: Only mutations in reproductive cells can be passed to offspring.

GED-SCI4-027 — Correct: C — Correct Answer: 4°C/min — Explanation: Temperature increased by 36°C over 9 minutes. The rate is $36 \div 9 = 4^\circ\text{C}/\text{min}$.

GED-SCI4-028 — Correct: A — Correct Answer: Day and night — Explanation: Earth's rotation causes locations to move into and out of sunlight, producing day and night.

GED-SCI4-029 — Correct: A — Correct Answer: Glucose is a fuel source used to release energy. — Explanation: Cells use glucose during cellular respiration to release usable energy.

GED-SCI4-030 — Correct: A — Correct Answer: More energy is needed to produce the same temperature increase in a larger mass of water. — Explanation: The table shows energy added increasing as water mass increases while temperature increase stays at 10°C.

GED-SCI4-031 — Correct: A,B — Correct Answer: Use the same exercise length before each pulse measurement. | Measure pulse for the same amount of time in each trial. — Explanation: Consistent exercise length and pulse-counting time make the results more reliable.

GED-SCI4-032 — Correct: B — Correct Answer: Rope B — Explanation: Speed = distance ÷ time. Rope B has speed $20 \text{ m} \div 4 \text{ s} = 5 \text{ m/s}$, the highest value listed.

GED-SCI4-033 — Correct: A — Correct Answer: Decomposers — Explanation: Fungi break down dead organisms and return nutrients to the soil.

GED-SCI4-034 — Correct: A — Correct Answer: Vaccination can prepare the immune system for a faster future response. — Explanation: Memory cells help the immune system respond more quickly if the real virus is encountered later.